The University of Waikato: Studio Report 1998-2001

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Abstract: *This studio report from Waikato University updates a previous one presented at ACMA97 (Whalley 1997). Covering the period 1998-2001, it outlines current developments and directions; research, composition, and collaborations; CD production; international administrative contributions; and some current speculations.*

1. Introduction: Studios & Campus

The University of Waikato first launched its major in Music in 1991, and now offers a professional BMus/MMus programme that extends to doctoral level.

The computer music studios were housed in two purpose built rooms in 1996. The two main rooms now include six sub-studios. In 2001, a multi-venue performing arts complex was opened on campus with a concert chamber wired to accommodate electroacoustic music interaction and broadcast. In 2002, a further studio room was added in the department with a suite of entry-level machines focusing on notation/composition and aural training.

The three-year undergraduate digital music programme (including Music and Computers, Composition for the Screen, and Electroacoustic Music) has now run for over ten years, supporting a graduate programme (studio based electroacoustic music, and live/interactive work) . Technically, the undergraduate programme covers sequencing and DA based composition, sample manipulation and synthesis generation, and work with MAX/MSP as well as generative systems. The undergraduate courses also contribute to a conjoint multimedia programme run by the Department of Computer Science. Graduate work is focused on the IRCAM suite of programs.

Ian Whalley established and directs the studio teaching and research programme. In 2000, Senior Tutor Lisa Meridan-Skipp assisted him, later appointed as lecturer and Studio Director at Victoria University of Wellington beginning in 2001. Composer Michael Williams took up the position from 2002. The current studio technician is Kim Walker, an accomplish percussionist with both music and computer science qualifications.

As the technology stream of the music programme has become a defining characteristic of the Department, we are fortunate to be well resourced with regular software and hardware upgrades, and can afford to be experimentally focused. The studios are designed to focus on digital music generation and composition, rather than recording.

The current studio configurations are on the website (www.waikato.ac.nz/music). They are supported by a 1Gig network connected to the Department's server. An advantage is that a studio can be individually configured for each user as they log on, and the 'studio' is in reality a network of computers that can be accessed both on and off campus.

An emerging issue facing us, as with many academic studios, is that hardware and technological literacy is becoming less a defining difference between professional and home studios. The differentiation is now largely made through the range of software used, the exploration of non-commercial aesthetics, and the development of skills to explore new composition strategies the technology affords. Artistic outcomes, enquiry and discussion then remain the primary focus.

Waikato University was the original gateway for the internet into New Zealand, and we are fortunate to maintain a first rate digital infrastructure.

A further noted international strength of the campus is its Computer Science Department, involved in a wide range of government funded and commercial research projects, and responsible for various spin-off companies. Computer music remains a strong interest in the Department and it makes ongoing international contributions. Much of this work centred on the output of Ian Witten (machine learning, digital libraries [Witten 1999]), David Bainbridge (optical musical recognition music information retrieval [Bainbridge 1997],) and the now departed Lloyd Smith (artificial intelligence applications and web based melody indexing services).

2. Studio Research, Contributions& Collaborations

2.1 Research and Composition

Studio research has centred on the interests of the director, tutors, and collaborators. Output covers a wide gambit, and has been extensively published internationally.

Digital music aesthetics (Whalley 1998b, 1999c, 2000m, 2001e) as an aspect of the problems faced by computer musicians is a continuing interest, as are issues of technology facing contemporary composers, such as the application of physical modelling synthesis or web-based music distribution (Whalley 1998c, 2000b, 2001c).

An interest has been in generative systems in music (Whalley 1999a&b) particularly in the application of system dynamics modelling as a way of simulating and generating new works.

This work formed the basis of an interest in building system dynamic models of emotional/cognitive response in computer music (Whalley 2000a&e, 2001g) and my being invited to lead the cognition workshop at ICMC2000 in Berlin (Whalley 2000d).

The 0-0 approach to cognition using system dynamics software afforded the development of a number of models that could be used as the basis for a closed system non-linear composition system, embedded in interactive artworks. Further work in this area is continuing, and the University has now registered a provisional patent over the prototype developments (Whalley 2001f).

Composition remains at the heart of the studio, and the large number of students passing through the studios contribute to a growing body of work contributed to local and international events. Increasingly, students are now producing works for mix media, and for live interactive performances. Briefly, recent staff contributions include *Ga no mita mono* (Whalley 2000c) written at Kunitachi's Centre for Computer Music and Music Technology; IZUII (Whalley 1998) premiered at ACMA; and Lissa Meridan's *twitter tourniquet* was released on CD in 2001 (Meridan 2001).

2.2 Reviews & Professional Publications

Reviews and professional documentation is part of the ongoing work associated with the studio. Over the period I wrote extensive book, software and CD reviews for a range of publications such as *Array Live* (ICMA), *The Computer Music Journal* (MIT), *Music In New Zealand*, *Contexts* (AUST). A sample of these from 2000 is given in the reference section (Whalley 2000 h-l). Although time consuming (for example a recent CD review of 'Southern Cones' from *Leonardo Music Journal* took four days), it provides a way of keeping abreast of the range of material being released, and integrating our activities back into the international circuit.

Documenting the work of other New Zealand composers involved in computer music is also ongoing. Recent contributions include articles on Kit Powell, now resident in Switzerland, and Matthew Suttor, now resident in the United States (Whalley 2000f, 2001b).

2.3 Visitors and Collaborations

Recent studio visitors who gave lectures and interviews included Warren Burt in 2001 (Meridan-Skipp 2001).

Outward visits independent of conference and festival contributions, included my being Composer-in-residence at Kunitachi's International Centre of Computer Music and Music Technology (Tokyo 1998), Visiting Fellow in the Computer Science Department at Meiji University (Tokyo 2001), and awarded a York/Waikato Visitors in 2001 (yet to be completed).

The visit to Meiji allowed the continuation of work in generative systems through applying artificial intelligence programmes to non-linear models of music cognition.

The Waikato Campus recently launched the award winning graphics and multi-media degree programme under the direction of Ian Gwilt. This is based on the conjoint programme run for some time in conjunction with Wanganui Polytechnic. The cluster in Digital Arts combining our studio, computer science, and computer graphics affords the ongoing development of a teaching and research programme where there is an alignment of international expertise (Whalley 2001o). This recently allowed us to bid for a Centre of Excellence grant as part of the Waikato (host) Auckland/Auckland University of Technology Information and Communication Technology proposal. The development of the cluster is ongoing, and more staff with an interdisciplinary approach and skill have recently been hired to support the development.

3. CD Production

With the major studio upgrades that took place in 1999, the main studio was ideally suited to CD mastering.

Through Waikato University acting as publisher, I led the New Zealand Sonic Art CD series initiative as editor and producer. The first call for works for the first CD went out in 1999, resulting in the *New Zealand Sonic Art 2000* (Whalley 2000d). This included nine composers, and aimed at updating the series of electroacoustic works last released by Ode Record Company in 1994.

There were far more submissions than space allowed, and the CD attempted to represent some of the diversity of work being undertaken, and balance new with established composers.

New Zealand Sonic Art Vol.II (Whalley 2001d) included six composers and was released the following year, dedicated to the memory of the country's electroacoustic music pioneer Douglas Lilburn who recently passed away. The dedication restricted the outcomes somewhat to a 'tape pieces' approach, with many of the works being more reflective than the first CD.

Both discs have been widely internationally distributed, and received many positive reviews from commercial sources such as *The Wire* UK, and academic sources such as SAN UK, and EMF. Both CDs also made the New Zealand Concert FM listening charts, finding a wider audience for the content outside the traditional computer music conference circuit.

A call for works for the third CD was released early in 2002, with the aim of completing the master by the end of the year. The focus of this third CD is slightly different than the first two, so as to encourage a wider variety of artists to submit material, use more local source material, and move beyond the approach dominant in the first two CDs. I hope to include works from Maori composers and include live instruments with soundscape processing more typical of the upper North Island aesthetic.

4. Administrative Contributions

In 1999 the studios took charge of administering the web site for ACMA. This involved registering acma.asn.au in Australia, redesigning the site, and mounting it on local servers. Electronic versions of *Chroma* were also added. The site averages around 180 hits a month, with most interest being in downloading *Chroma*. Hits increase greatly when the annual conference is announced.

In the same year we formed a new listserv group (acma-l) as a service to the Australasian Computer Music Community. This incorporated the oz-computermusic and New Zealand Sonic Art lists. The list currently has around 140 members, quite in excess of current ACMA membership. Traffic is not heavy. Perhaps current state of government support for computer music in Australia is partly responsible for this ?

Other regional activities involve acting on the reading panel for *Micropolyphonie*, and acting as the regional editor for *Organised Sound* (Cambridge University Press).

Making and administrative contribution to computer music internationally has been an ongoing process. For example, in 2000 I sat on the final paper selection committee for the International Computer Music Association Conference in Berlin. The panel for took more than a month of work, selecting 133 out of 375 papers from international field. The process was a difficult one because of the high level of most work.

In 2001 I co-ordinated an issue of *Organised Sound* focusing on Computer Music in Australasia/South East Asia (Whalley 2001a). The request for submissions began by noting that along

with increasing globalisation, each world region has its distinct characteristics, and that Australasia/South East Asia represents an exciting crossroads of established, traditional and new approaches. The questions authors were asked to address included: Are there distinct national or regional aesthetics in electroacoustic and computer music in this corner of the globe? Which developments in relevant areas of music technology are taking place or are about to take place in Australasia/South East Asia? Are they related to distinct aesthetic ideas, compositional procedures, technological applications, and/or technological developments? Are there environmental or cultural circumstances that influence computer music in this region?

Authors from different areas submitted summaries of developments, cultural/economic backgrounds, events and approaches, written either individually or collectively. They also dealt with wider thematic and national issues as they say fit.

Nine papers were selected: two from New Zealand, one from SEA, and six from Australia. The material chosen attempted to cover most of the region and the themes that arose, although some geographical areas of Australia are neglected through a lack of material or it being insufficiently addressed to the central themes. The responses were as diverse and the region itself, and the collection captures some of the similarities, contradictions and differences that one would expect to emerge in a complex, evolving and diverse region. (Whalley 2001a). For the Auckland/Waikato summary, see (Elmsly, Dart, Whalley 2001).

5. Speculations

The New Zealand Government recently released a new *Strategy for Tertiary Education 2002-2007*, signalling its intention to focus university activity on research outcomes with links to industry to underpin its move towards a 'knowledge economy'. This will require greater networking and collaboration between universities, recognising each other's recent unique 'profile' and expertise: a change from the competitive model based on funding undergraduate student numbers. Within this new context, computer music/electroacoustic music can find a good niche nationally, integrated into the international computer music community.

Given the current developments on Campus, and the increasing integration of digital arts, our programme is likely to become increasingly web based, non-linear, and multimedia based, in tandem with increasing specialisation in computer music composition, interaction, and research.

Increasing government support for the creative industries also is likely to see the use of more local source material in products, a path more of our ex-students are discovering with increasing success outside academic music. This is also a part and reflection New Zealand beginning to finding an international artistic niche in a variety of new artistic media.

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